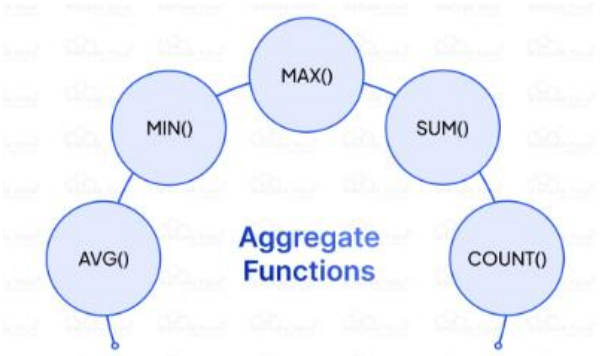
DBMS- SALES AND CUSTOMER ANALYTICS SYSTEM

Pepperfry Furniture E-Commerce Database Management System

1. Aggregate Functions

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single value as the result.



1.1 COUNT()

The COUNT() function is used to count the total number of orders available in the Orders table

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
          10
```

1.2 SUM()

The SUM() function is used to calculate the total sales amount by adding all the order amounts in the Orders table.

```
SQL> SELECT SUM(Total_Amount) AS Total_Purchase_Amount
  2  FROM Orders;

TOTAL_PURCHASE_AMOUNT
---------------------
               254500
```

1.3 AVG()

The AVG() function is used to calculate the average order amount by finding the mean value of all orders.

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;

AVERAGE_ORDER_AMOUNT
--------------------
               25450
```

1.4 MIN()

The MIN() function is used to identify the lowest order amount in the Orders table

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order_Amount
  2  FROM Orders;

MINIMUM_ORDER_AMOUNT
--------------------
               12500
```

1.5 MAX()

The MAX() function is used to identify the highest order amount in the Orders table.

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order_Amount
  2  FROM Orders;

MAXIMUM_ORDER_AMOUNT
--------------------
               45000
```

2. Generate Total Sales Report

This report provides an overall summary of business sales by displaying the total number of orders, total sales amount, average sales, minimum sales and maximum sales. It helps to understand the overall sales performance of Pepperfry

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Sales,
  5      MIN(Total_Amount) AS Minimum_Sales,
  6      MAX(Total_Amount) AS Maximum_Sales
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES MINIMUM_SALES MAXIMUM_SALES
------------ ----------- ------------- ------------- -------------
          10      254500         25450         12500         45000
```

2.1 Customer-wise Sales Report

This report displays the total number of orders and total purchase amount for each customer. It helps to analyse individual customer purchasing activities and understand customer sales contributions.

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      COUNT(O.Order_ID) AS Total_Orders,
  5      SUM(O.Total_Amount) AS Total_Purchase_Amount
  6  FROM Customer C
  7  INNER JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_PURCHASE_AMOUNT
---------------------
          1 Arun                                                          1
                25000

          2 Priya                                                         1
                18500

          3 Rahul                                                         1
                32000


CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_PURCHASE_AMOUNT
---------------------
          4 Sneha                                                         1
                12500

          5 Karthik                                                       1
```

3 . Find Top Customers Based on Purchase Amount

This report identifies the customers who spend the highest amount on furniture purchases. It displays customers in descending order based on their total purchase amount

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS Total_Purchase_Amount
  5  FROM Customer C
  6  INNER JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name
  9  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          5 Karthik
                45000

          9 Ajay
                37500

          3 Rahul
                32000


CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          7 Vijay
                28000

          1 Arun
                25000
```

4. Identify Best-Selling Products

This report identifies the most popular furniture products based on the total quantity sold. It helps to understand customer preferences and identify products with higher demand

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OD.Quantity) AS Total_Quantity_Sold
  5  FROM Product P
  6  INNER JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
         1
Sofa
                   4

         5
Bookshelf
                   3

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

         9
Study Table
```

4.1 Product Sales Amount

This report calculates the total sales amount generated by each product using its quantity sold and unit price. It helps to identify which products contribute more revenue to the business.

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OD.Quantity * OD.Unit_Price) AS Total_Sales_Amount
  5  FROM Product P
  6  INNER JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Total_Sales_Amount DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_SALES_AMOUNT
------------------
         1
Sofa
             50000

         5
Bookshelf
             45000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_SALES_AMOUNT
------------------

         9
Study Table
```

5 . Category-wise Sales Analysis

This report analyses the sales performance of different furniture categories. It displays the total quantity sold and total sales amount for each category, helping to identify the most profitable product categories.

```
SQL> SELECT
  2      C.Category_ID,
  3      C.Category_Name,
  4      SUM(OD.Quantity) AS Total_Quantity_Sold,
  5      SUM(OD.Quantity * OD.Unit_Price) AS Total_Sales_Amount
  6  FROM Category C
  7  INNER JOIN Product P
  8  ON C.Category_ID = P.Category_ID
  9  INNER JOIN Order_Details OD
 10  ON P.Product_ID = OD.Product_ID
 11  GROUP BY C.Category_ID, C.Category_Name
 12  ORDER BY Total_Sales_Amount DESC;

CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
         3 Chair
                   3              51500

         1 Sofa
                   4              50000

         5 Storage
                   3              45000


CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
         7 Dining Table
                   2              28000
```

6. Combined Business Sales Report

This report provides a complete overview of Pepperfry's business performance in a single report. It displays total orders, total sales, average order amount, minimum order amount, maximum order amount and total quantity of products sold. It helps to evaluate overall business performance and supports better business decisions.

```
SQL> SELECT
  2      (SELECT COUNT(*) FROM Orders) AS Total_Orders,
  3      (SELECT SUM(Total_Amount) FROM Orders) AS Total_Sales,
  4      (SELECT AVG(Total_Amount) FROM Orders) AS Average_Order_Amount,
  5      (SELECT MIN(Total_Amount) FROM Orders) AS Minimum_Order_Amount,
  6      (SELECT MAX(Total_Amount) FROM Orders) AS Maximum_Order_Amount,
  7      (SELECT SUM(Quantity) FROM Order_Details) AS Total_Quantity_Sold
  8  FROM DUAL;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_AMOUNT MINIMUM_ORDER_AMOUNT
------------ ----------- -------------------- --------------------
MAXIMUM_ORDER_AMOUNT TOTAL_QUANTITY_SOLD
-------------------- -------------------
          10      254500                25450                12500
               45000                 20
```